

1

11.11.2023

Software engineering

lecture # 01 After Mids

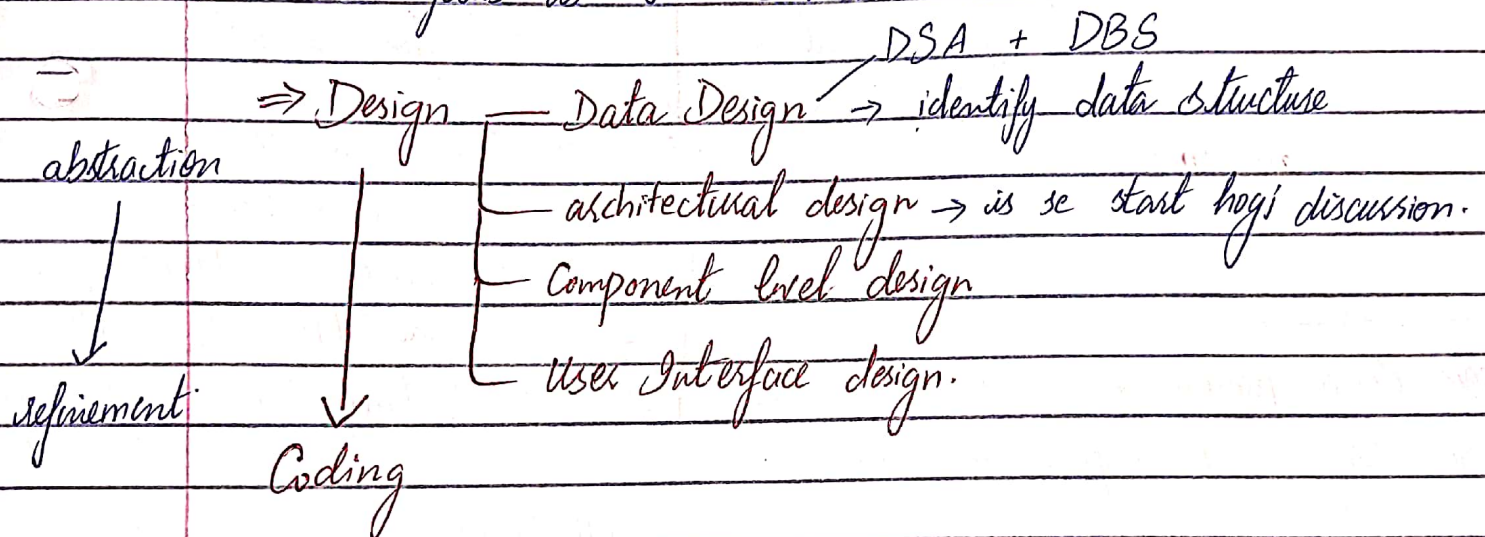
6-Nov-2023

Monday

Design Concept & Principles.

In analysis $\rightarrow$ our focus is on 'what' $\rightarrow$ Data modeling
Functional "
Behavioural "

$\Rightarrow$ Design $\rightarrow$ "How" (Solution) $\rightarrow$ on the basis of what
Total focus is on solution now



$\Rightarrow$ Fundamental Concepts

1) Abstraction, refinement

eg World-cup $\rightarrow$ It is difficult for Pakistan to reach semi-finals ^{1st} and ^{2nd}

abstraction $\downarrow$ refinement $\uparrow$ $\leftarrow$ aisi kosi conditions qis se semi-finals mai ja sakte.

jahan details zyada waham refinement zyada.

Abstraction is high level view of anything

In analysis decomposition → before decomposition abstraction zyada hoti hai but after abstraction kam.

Divide & Conquer → cheezen decompose karna yani refinement zyada se zyada karna.
Abstraction & Refinement

There are ways to go from abstraction to refinement but these ways depends on things-

Types of Abstraction — Data Abstraction → eg data ko high level se low level par ly jana - sab kuch data se related of dm ka data.

algorithm

Procedural "
Behavioural "

is mai sara procedure of kam chalta kaise hai - data k behaar se details mai chly gae toh uski data ki abstraction kam ho jaye gi

Logic dena programmer ka kaam nahi hai ye designer ki responsibility hai

"Right Man at Right Time for Right Task"

Abstraction — Refinement
divide & conquer.
↓
is se result acha ata hai-

2) Modularity :- (division karne ki degree for abstraction & refinement)

Component → RAD
or module
→ independent Functionality
→ appropriate interface

Deom
FCIT ki modularity?



These all are modules.



2 Organizational Chart (Organogram)

4 modules
lazmi ayen
gy

① input k modules.

② output k modules.

③ processing modules (jinhe me kaam kar k dena hai)

④ Decision makers

↳ condition.

information sharing is not the dependency (yami agar 2 logo k darmyan koi information share ho sahi ho toh ~~is~~ not necessary k dependency bhi ho). → yami in

resources best optimal level pay use ho - mai sequence matter nahi karta -

agar koi cheez humary samne divide krni hai toh uski division kerty level tak jaye gi.

* Control Hierarchy and

in this case.

modules kis level tak bnayen gy-

→ Program Structure

cost → integration cost → modules k zyada hony se ye zyada hogi -
↳ development cost → without module.

↳ and

→ call &

jahan dono cost equal hojaye → optimum point

Return Architecture

has aik ki zimmedari declare hoti hai - k kis ne kya karna hai -

In number

Depth → vertical levels (total) (main se start krna hai) -
(4)

Breadth → maximum → jitni divide hongy → horizontal -

Executive ka super ordinate module $\rightarrow$ none

upar waly
sub ordinates $\rightarrow 17$
 $\downarrow$
neech waly.

This is the control hierarchy.

super ordinate
& sub ordinate modules.

communication channel
executive se through
maintain hoty hai

From In From Out

$\swarrow$ module
M or N
super ordinate
multiple or
only one
sub ordinate
yani-multiple
ko aik deal
kar saka hai

$\swarrow$ multiple
sub ordinates ho
or
aik exactly aik
super ordinate ho
then From out

$\rightarrow$ This is the
normal position

$\rightarrow$ Natural situation.

So naturally hierarchy
bonaty hue ham ne
From Out py rehna hai

eg ham sir se
sub ordinate
hai
or sir binary
aik super
ordinate then
From out

$\rightarrow$ This is not a normal
situation (abnormal)

eg CMS Login (Login Credentials) $\rightarrow$ agar aik hi id log use karin
sharing karin then lagayi hoti hai $\rightarrow$ yani From In.

(B) Total modules $\rightarrow 17$

1 $\rightarrow 17$
2 $\rightarrow 17$

$$17 \times 17 = \frac{n(n-1)}{2}$$

Formulae.

$$\begin{aligned} \text{For modules } & \frac{17(17-1)}{2} = \frac{17(16)}{2} = \frac{10(10-1)}{2} = \frac{10(9)}{2} = \frac{90}{2} \\ & = \frac{272}{2} = 136 \text{ total communication channels} \end{aligned}$$

= 45

①

Software Engineering
Lecture 17

Software Engineering

Lecture # 02.

Wednesday

8-Nov-2023

Visibility

Connectivity

↓
some like Public, Private
& Protected (Specifiers) in
OOP-

hum specify karte hai ki
kisi ki visibility ya access
kahan tak ho-

if private → then usi
body tak usi
ki access

In program → main
function puri program
ki access rkhata hai-
lekin jo main ke
andar hoga ~~uski~~ uski
kam hogi-

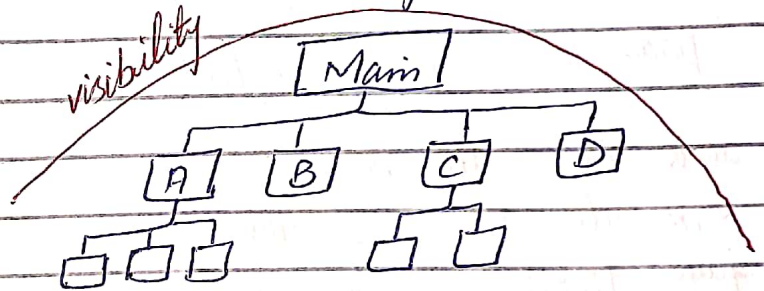
jab kisi dusri ki
visibility mai interfere
kary ga → toh
problems ayen gi-

↓
interference

visibility means dekhna
ap ka kisi kahan tak
chale dia hai

→ visibility ke andar connectivity
chal sakti hoti hai-
umbrella ki visibility hai-

→ aik particular module direct
dusre modules ke sath
connected hai
lekin visibility kahan tak hai-



ab main kis ke sath connected
hai direct

↳ that is connectivity.

eg Deem our FCIT
visibility sab ke sath hai
but direct connect kuch
log hoty hain.

Call of Function → connection

generate
ta ke wo execute ← karti hai
ho → is sab ko control
main karta hai-

connections → lines → connectivity or jo scope specifies hoty hai they define visibility
har aik module ki visibility bhi chal rahi hoti hai plus connectivity bhi-

eg Agar Sir Ejaz ko IT ki degree mai koi kami hai toh Sir direct IT ki degree supervisor se baat nahi karein gy → balky deen ki paas jayen gi → connectivity

agar aik sath hi 2 modules ko access mil jaye then program crash ho jata hai-

⇒ How to maintain this & manage this?

→ Functional Independence -

opposite → dependence

self sufficiency
خود کفالت

⇒ Information Hiding
same as encapsulation
"Ignorance is bliss"

غیر ضروری چیزوں کا بتانا نہ لینا

eg apko PM کی اختیارات کی mil jayen
ya Deen کی اختیارات کی mil jayen -

غیر ضروری کا بتانا نہ لینا

Software mai wohi karna hai jo zaruri ho-

User jesa marzi ho lekin job wo access kary toh us ko lagy ki us

jo zaruri cheezin hai ye software ho wo show karna di jayen -
eg access specifier.

② سوچنے کی آزادی → Freedom to think

③ فیصلہ سازی → Decision

④ عمل کرنے کی آزادی → Action

ہمارے پاس ہے۔ ہفتوں آزادیاں → ham inko use kaise karty hain → یہ امتیاز ہے

Functional Independence

① Association ② dependence

jo kaam independently ho rahi ho.

every din k kaam jo ham ne kiye eg nashta, ride li, uni agar → so ye sab kaam connection hai → association hai → dependency nahi hai.

eg hmari aik din k sab lectures they are associated with each other but not dependent.

These are some students jo apna nashta khud bnaty ho → wo independent hain or some jo apny parents par dependent hoty hai jo nashta or kahnin se ly kar karna hai toh you are dependent on them.

Association is better.

Types ① Cohesion → aik module khud apna kaam karna koe leta hai - us hisaab se degree-

② Coupling → agar pure association ho toh coupling ki degree 0 hogi

jab ham ~~decision~~ design ya architecture bnaty hai-
toh aise module ka dekhna hai k wo apny sab
kaam khud kis degree par kar raha hai-
or agar wo isi k sath dependent hai-

agar cohesion 100% ho $\rightarrow$ yani wo apna sab
coupling 0 $\rightarrow$ no dependency kaam khud karta.
hai

They are reverse of each other. $\downarrow$ only association hogi

Principle of functional independency:-

"While developing the design we try to maintain low
coupling and high cohesion"

aisa design ya architecture jis mai zyada tar
kaam khud kia jaye-

low coupled design $\rightarrow$ our priority

$\Rightarrow$ highly cohesive low coupled
jab logic bana rahi ho-

$\rightarrow$ input
 $\rightarrow$ output
 $\rightarrow$ Processing
 $\rightarrow$ decision

$\Rightarrow$ Types of cohesion & coupling (jo ham during programming
apply karte hai).

$\Rightarrow$ materialistic
challenge

or خورشید

3 cheezin

① apna ghar (own house)

② Ride (سیارے)

③ Spouse (Good
Life Partner)

Challenge by Sir $\rightarrow$ jab ap yahan se free hojayein gy 2025 mai
tab degree mily gi $\rightarrow$ 2030 mai dekhna hai $\rightarrow$ kya
ye 3 cheezin hmare paas hai- agar hai toh you are success

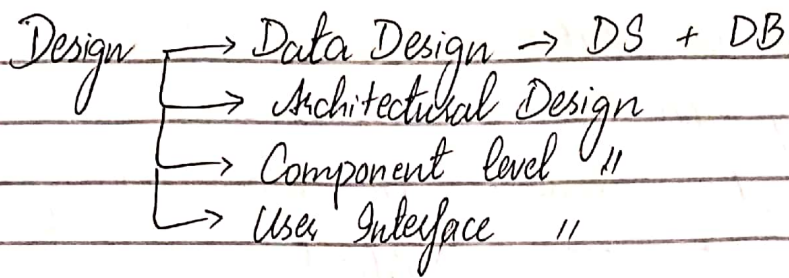
①

13-Nov-2023

Lecture # 03
After mids SE

①

Monday



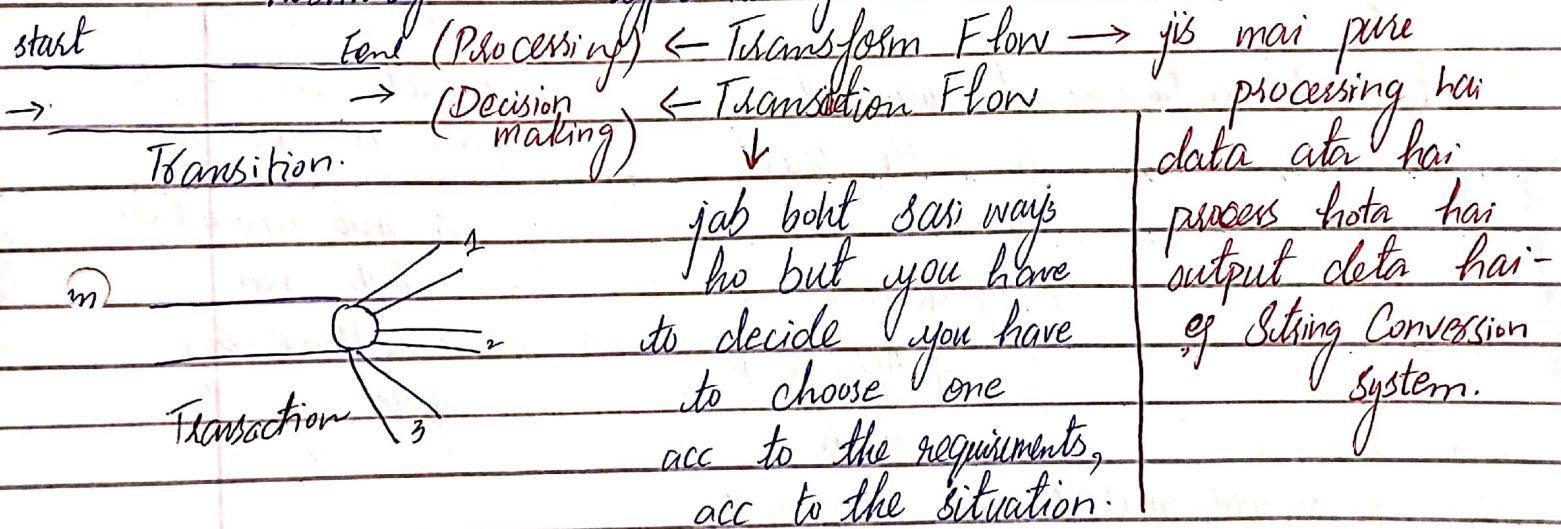
Architectural Design

→ Architecture → Placement of the modules & components according to the requirements.

Control Hierarchy is an example of architecture.

How we can abstract the architecture from analysis model?

DFD mai data ka jo flow hota hai wo normally two types ka flow ho skta hai.



validation is a process but after this we have to decide so it is a combination of both.

we have combined them so low cohesion high coupling

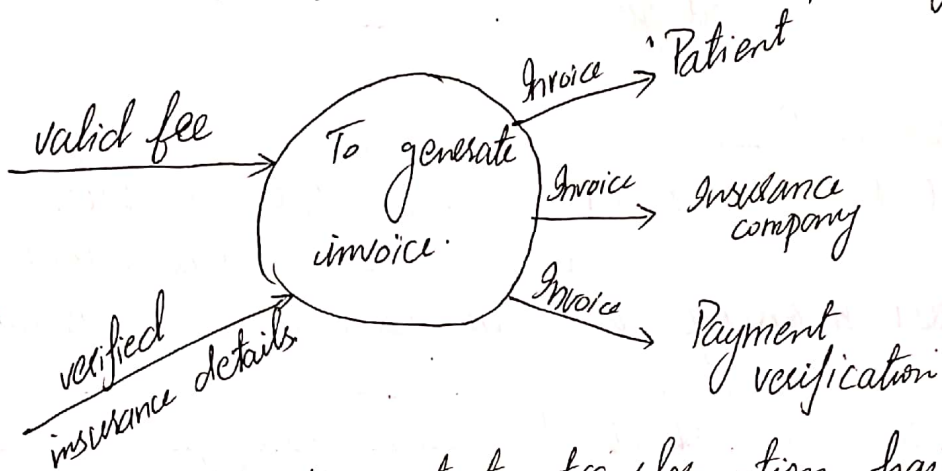
⇒ effective / productive / relative decision making.

situation should be clear $\rightarrow$ then apka decision sahi sabit hoga.
but if you are not clear then decision sahi sabit nahi hoga.

2 input $\rightarrow$ receive kar raha

output $\rightarrow$ 3 dy raha hai-

but at a time aik hi output aye ga-



ye output transformation hai $\rightarrow$ is mai same output

aik hi time par related entities k paas ja skti hai-

is mai association hoti hai simultaneously jaye gi- $\rightarrow$

but in transaction $\rightarrow$ aik output ka hi path select hoga:-

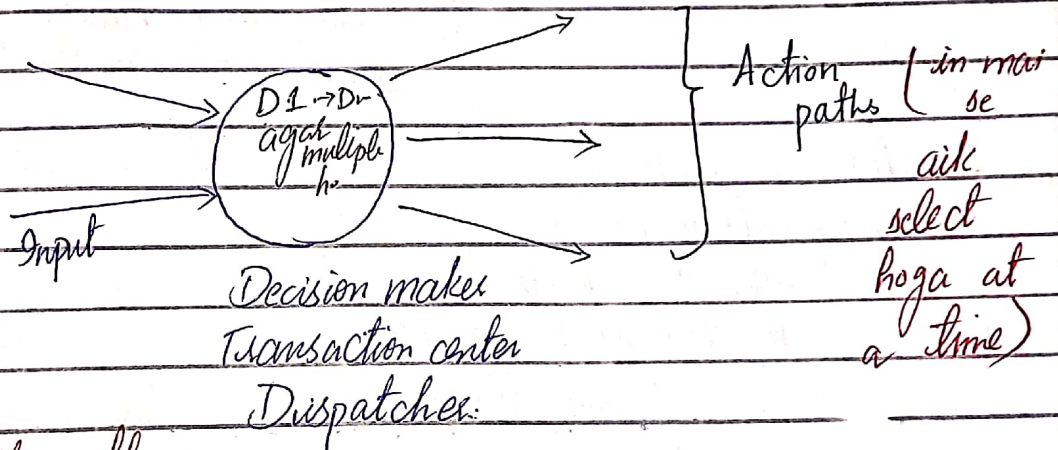
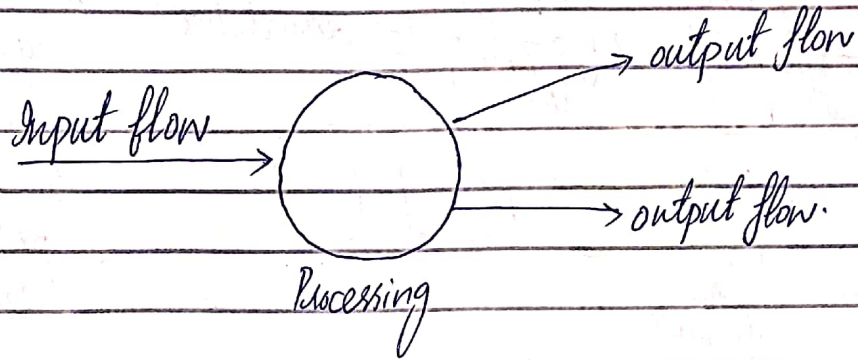
In Programming
if/else + switch

is mai multiple path hongy, you have to select the related path.

Traffic signal or warden

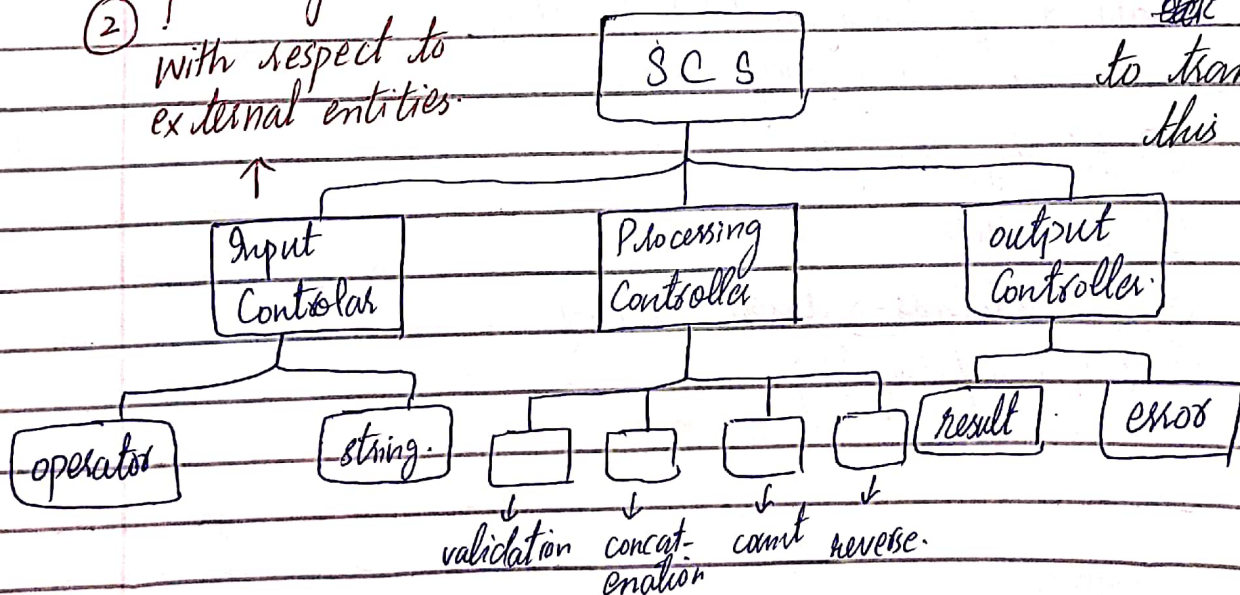
$\rightarrow$ only pure transaction
aik time py aik hi jaga-

- (2) Pure transformation → only Processing → aik hi output multiple related entities k qas j shta hai -
- Pure transaction → only decision making, only one path only one output at a time.
- Both (transaction + transformation)



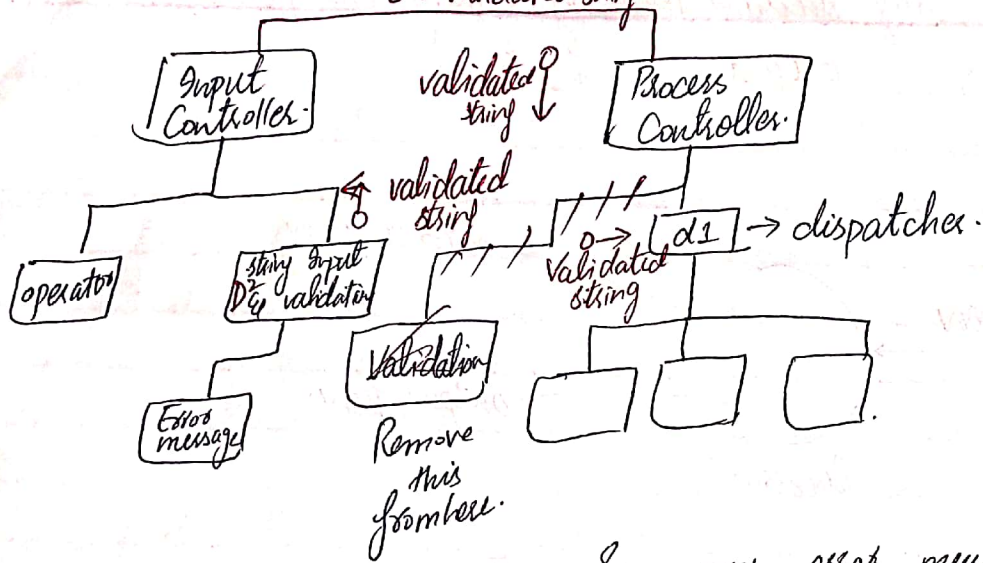
- ① Identify the flows
↳ 1st step in architecture.

- ② transformation k acc modules.



With respect to transformation this is the architecture

Now we have to involve decision making.
 dispatcher → action path se pehly.
 → validated string



us sab se coupling zyada ho rahi hai.
 So we have to follow ~~this~~ design principles in order to increase cohesion.

So now error message output se remove ho jaye ga-

Captcha in CMS

→ frontend py hai
 user k behaaz se run time py ki jati hai-
 coz of non-functional requirements.

~~width~~
 breadth → 4
 depth → 4

⇒ Mapping of transformation is easy then decision making

- ① Identify the flows
- ② Controller
- ③ Dispatchers → decision making
- ④ Cohesion / Coupling.

(1)

SE

25
24
34

15-Nov-2023

Lecture # 04

Wednesday

Component - Level Design

modules ki logic abstraction se refinement ki taraf jana -

har aik module ko individually deal karna hai

If we have 40 modules then 40 component-level-design.

"Component level design is considered the closest activity to coding"

we have some options to define logic:-

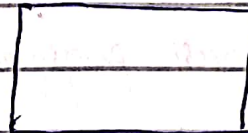
- graphical (eg flow chart)
- pseudocode
- programming language.
- decision table
- conduct walkthrough

codes ka kaam sirf syntax provide karna reh jata hai agar software engineers ko kuch ghalat lag raha ho toh wo report kar sakte hai but change nahi-

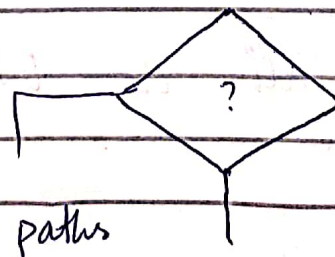
logic ki 3 types ki statements hoti hai

- 1- Sequence statement → jab us statement ki basi
eg declaration, assignment of a variable, return call etc
aye to wo execute ho ke call next line ko dy

This is shown by



2- Conditional Statements



3- Repetational Statements. No ~~if~~ symbol for this
 loops etc arrow head se maintain kia jata
 hai-
 arrow head se adjust kia jata hai
 is ko ...

* These 3 are programming constructs.

Note:- to show the process (use of ^{Flow Chart} ~~component level design~~)
 بنانا جس کی Process کو show کرنے کے لیے سب سے پہلے
 Flow Chart ko use kia jata hai-

Pseudocode (eg PDL) ^{Program Design Language}
 Programming language
 Algorithm → jis cheez mai bas logic ho.
 ↓
 loose structure. steps hongy or logic hongy.
 independent of the programming language.
 mini. specification algorithm
 hoti hai-

Stepwise Refinement
 slide. 319

2nd step → ~~pseudocode~~ algorithm

1st step → Process.

3rd step → Pseudocode.

(PDL) Pseudocode → the mixture of logic & syntax of Programming language.
 → is mai Programming language ka syntax hona zaruri hai-

Programming language. → sari programming syntax

These all are toolkit (jab kaam karna hai toh kisi
 aik cheez ko use karna
 hai)

⑦

aik support ham use krty hai.

⇒ Decision table

If (OR AND AND)

$C_1 \qquad C_2 \qquad C_3 \qquad C_4$

If (C_1 & C_2)

Permutations $\rightarrow 2^n, 2^3 = 8$

When both conditions is not true $\rightarrow$ don't case situation

Delegate $\rightarrow$ kisi se kaam krna lena.

account transfer krni hai

1
5

B → key in PP

SE

Lecture 20

Date: 20/11/2022

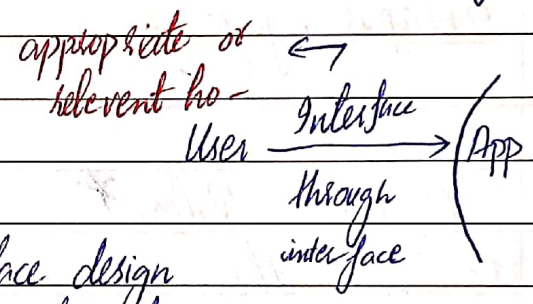
Lecture # 05

Monday

jitni marzi achi ← User Interface Design → last portion of the logic bana lein structural design.
agar user interface interface kis tacha ka hona chahye?
acha na ho toh that is the problem.
logic kisi kaam ka nahi Implementation is not the problem.

eg You have car → lekin ap ko chalani nahi ati.
ap driver rakhty hai → you are creating interface

Interface is the communication medium, a usage medium



eg cms ajkal use nahi ho raha because of captcha → interface design theek nahi hai.

↓
A Control Problem → application pay ap ka control nahi hai.
① Response time problem. (load home mai time ly rahi hai)
② Placement of Content / Controls.

I am controlling the application not application controlling me.

well classify nahi hai
Good Practice is abstraction
organize nahi ha → aik hi page py boht sara data ho.

↓
Refinement.

change jab karna ho toh uska asol hai k fit abruptly change na ja jaye → lekin ahista ahista layen gy toh that is acceptable.

B - Consistency

means user ka control hai.

Date:/...../20.....

Abstraction → refinement,
decomposition
divide & conquer.

So is k according consistency ka concept ala hai.
eg word, excel, ppt → basic layout same hai inka.
consistency honi chahiye.

C - Memory Load → kaam karthy hue cheezon ko
yaad rkha parta hai-

In calculator

↳ M+

↳ memory store karta hai-

M-

↳ wapis a jaye ga

to
(reduce the memory load of
the user.)

In word → tooltip

help ka menu

Ctrl + C

Shortcut keys.

Excel → mai
calculation

Pace Special

Costom ka use.

Control ka

nam common
life ka metaphos
ko-

Golden Rules:- *

backend

py ye

chalti

— hui nazar
ani chahiye.

- 1- Place the user in control
- 2- Reduce the user's memory load
- 3- Make the interface consistent.

⇒ jaise coupling ko kam
cohesion ko zyada.

VCR

Typical Design Errors.

- lack of consistency

- too much memorization

- no guidance / help

- no context sensitivity

- poor response

- Arcane / unfriendly.

consistency mai
hi ata hai

ke pehly kya

hua hai baad mai tyaar nahi → usko current ke sath link
kareni

پہلے سے جو چیزیں
ہیں انہیں
نئے سے جو چیزیں
ہیں انہیں
سے
کے ساتھ
link
کرنی

user name	task	environ
—	—	—
—	—	—
—	—	—

⇒ User Interface Design Process

(4)

① user, task & environment analysis

(sab se pehly ye identify karna hai-)

→ ye kaam pehly hua hai → just abstract karna hai-

→ user ka naam or us k from exd or dfd.

task

HCI

↓
Human
Computer
Interaction

↓
Interface
design ki
latest form.

② Interface design (identification of screen on basis of 1st step)

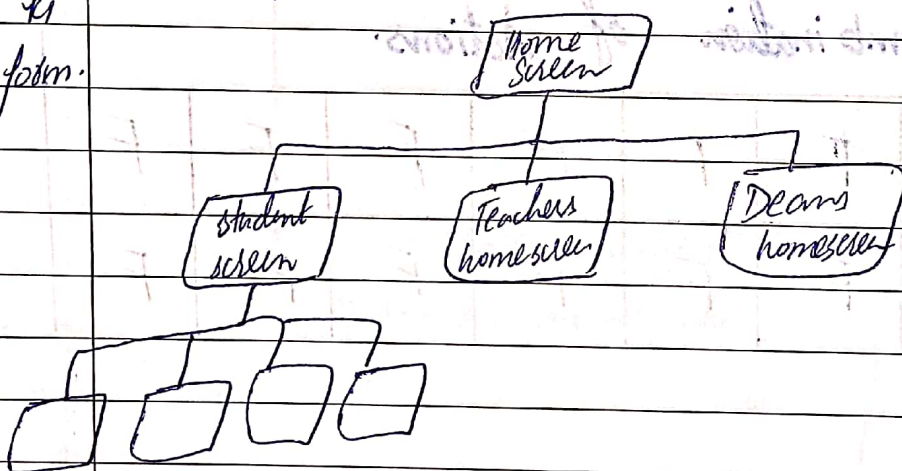
① 1st level pay ham hierarchy bnayen

gy screens ki-

usko sitemap kehety hai-

→ architecture ki form mai hi
banta hai-

of Cons



ye sara kaam
1st step ki
basis py ho raha
hoga-

kisi bhi website pay kisi corner pay sitemap dikhha hoga.
you have to check it

pehla kaam → sitemap

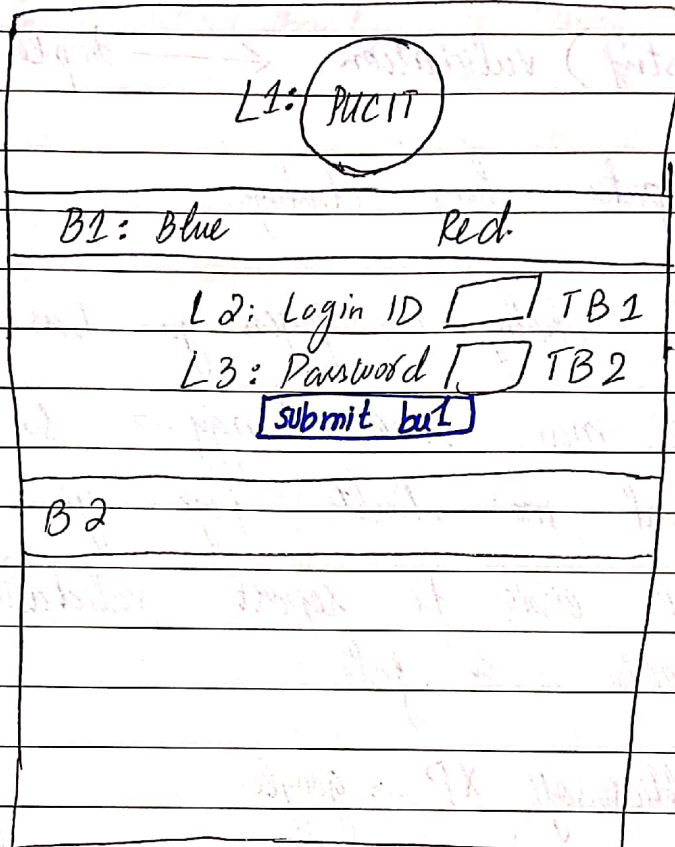
② Story Board → jitni screens hongy
utny hi story
board hongy.

35

(static content is known as label shown as L) eg monogram logo

Date:/...../20.....

→ Story Board



TB → Text Box

L1: This is label containing PUCIT

B1: Banner 1 contains the

L2:

L3:

TB1: Text Box takes the value of login ID

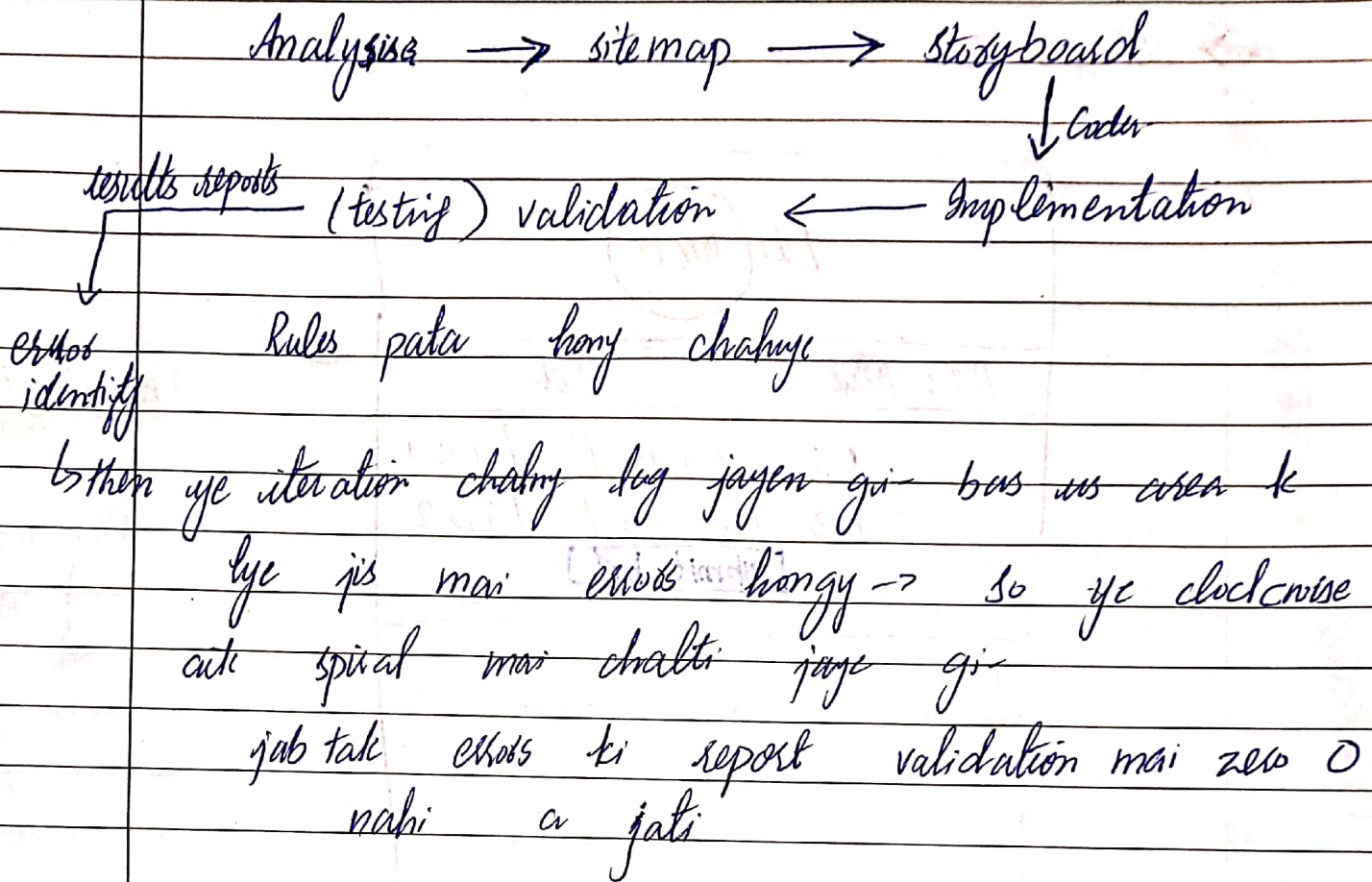
TB2:

BU1:

B2

Search
⇒ list of User Interface controls.

Date:/...../20.....



eg Microsoft XP → google

↓
check its interface.

Windows 11 mai bhi almost yahi use
ho raha hai.

end of Structured analysis &
design.

1

SEDietal ki book
Programming ki-
Wednesday

Date: 22/Nov/2023

Lecture # 06

Use Case Modeling

Next Portion → Object Oriented Analysis & Design.

* Starting mai software structured approach mai banti thi. ab 90-95% programming or development object oriented approach mai hoti hai-

function kaise define karly hai dono mai
eg calculator dono approaches mai banana pay-
eg data management

What was the problem in structured approach that object oriented approach invented?

only two things — data { inki interaction se kaam hota hai }
function

Solution of Problem
(Simultaneous control of data & functionality change tha)

Control of data & function
→ structure mai kaise control hota hai? or object oriented mai kese data or function in structured alag alag pay hui hain independent hai

jab programs ka data zyada hona start hoga
phir ye problem zyada ati hai-

In object oriented → har function or har data aik class ke sath associated hai-

class ka concept real life concept hai e.g. entities un k agay unka object bnta hai unka particular data hota hai → functions hoty hai-

aisa kaam behtar
hota hai-
↑

Date:/...../20.....

Why object oriented? → to ^{give} simultaneous control of data & functionality

If two students started talking to each other, they have compromised on data & function jo teacher ki taraf se milna tha-phir us kuch changes karein gy q k us k pass complete control hai over data & functionality.
(teacher control rakhty hain)

ham ne or us k mil k chhota hua class ko-
but agar student boat na many → means violate kya hain rules ko?
structured approach

no structure OO approach mai hai class or object ka-
this can give us the simultaneous control of data & function.

(OOA & D)

① Use Case Modeling
ye L→ High Level Use Case Diagram
2 } → Analysis level " " "
analysis }
mai } → Use Case Description (Documentation)

② Domain Model

ye 2 cheezin analysis mai ati hai then design start.

③ System sequence diagram.

④ Sequence diagram

⑤ Design Class Diagram (DGD)

ye 3 design mai ati hai

These all diagrams are essential.

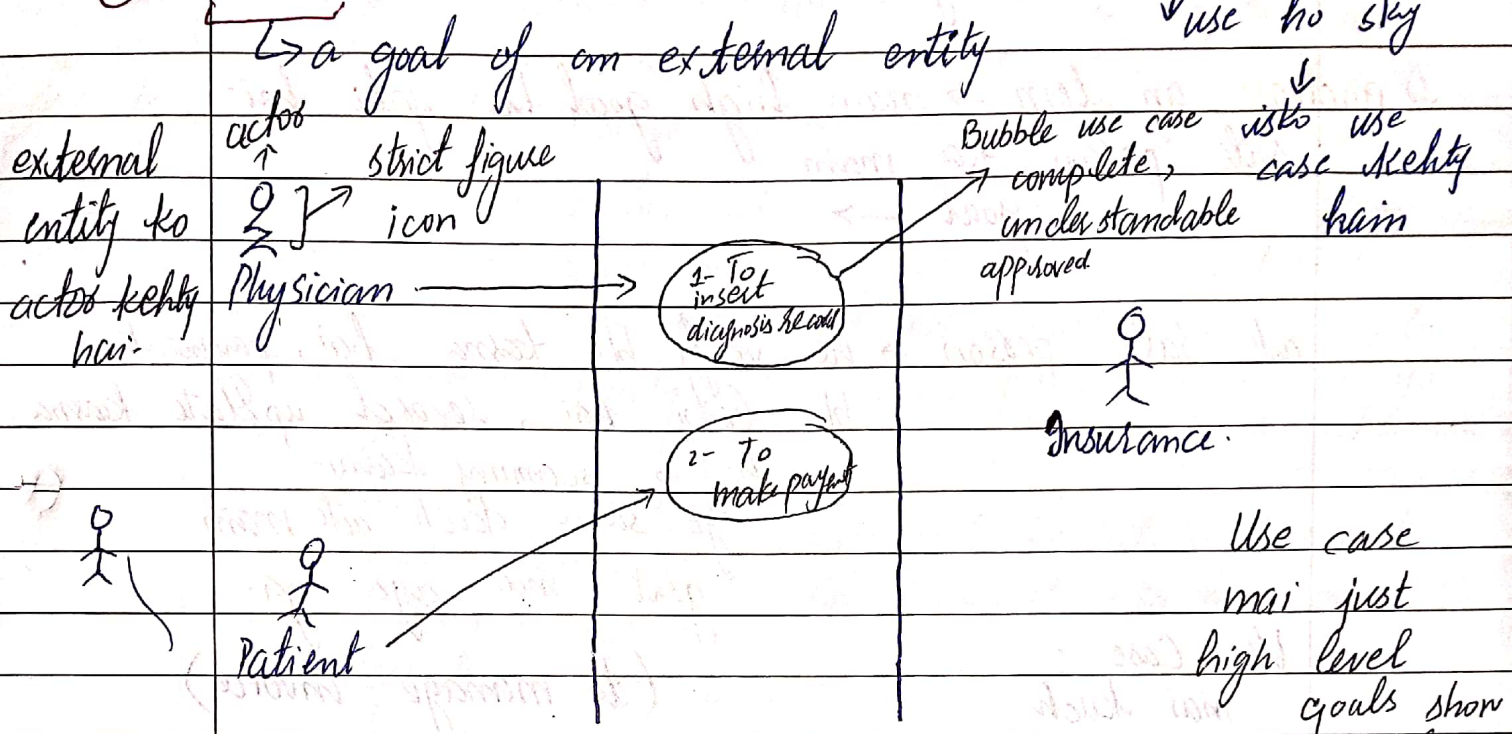
us actor hoga or name

2

recall structured → sab se pehly requir gather
ho k a gai hai ab unko
no description deni hai k analysis

Date:/...../20.....

① Use Case Model:-



1st we have to identify the external entities
ab in k sab k high level goal dekhne hai
(unka goal clear hona chahiye)

Invoice in teeno mai se kisi ka goal nahi
hai → wo inko chahiye but isko generate karna
uska goal nahi hai

Goals { Patient → insurance details or payment dena.
Physicians → diagnosis record dena.
Insurance → payment dena

→ our goal is to attain degree but is k sath boht
kuch chalta hai → then wo goal banata hai
aik actor hoga or uska complete understandable

Date:/...../20.....

use case hoga.

to purchase an item → main high goal bas yahi hai
sale person ka main goal →

aik sale person → ne verify bhi karna hai, payment
bhi leni hai, record update karna
hai, scanning karni-
ye sara kuch aik main
goal mai aye ga.

Use Case

mai kuch

change karein gy then
it will become



EBP (Elementary Business Process)

Student aik class hai



or specific student
object hai uska

Patient hai is ka goal hai
to make payment



jab in mai
values a gai toh ye

EBP ban gya.

Use case is the
class



EBP is the object
(It is actually the
physical existence of
use case)

Real Case execution
is mai values ati hai
other name → Concrete / Real
Use Case / Time
Use Case

Date:/...../20.....

Actor

→ 3 types → ① Primary actor → jis k lye system bnaya

② Secondary actor

③ of stage actor

↓
acc to
ODANot primary goal, no
generation support, bas
interested hoty info maijata hai jo directly
apny goal achieve
karn k lye system
k sath interact

acc to

CMS

Student
TeacherPrimary
Actorgoal ko position mai ya
sara hasil kar raha hota

Program Coordinator

Degree Coordinator

Account Officer

Parents

main goal ko achieve karn k
lye support provide karna.

↓

hamara result dekhna

in k main goal to

not their primary

jo support provide karen for

goal → na ye

support provide

Secondary goal.

kar sath generate

karn mai - They are

just interested in is se

hasil hony wali information

mai-

jo information generate krty

bas us mai unka interest

hota hai - of stage actor.

ye karn
waly of stage

↑ actors

CC

↓

Carbon
Copy
email
generate
krty
hueye just by concept
dekhna haidiagram mai show
nahi krty

Date:/...../20.....

After combining these all
use case mai 4 cheezin hon chahiye

① Goal (Particular, ~~truthful~~ complete, understandable)

shown by

② Actor (multiple
bhi ho sakti hai)



③ Scenario
(EBP) (Details of goal)



④ System (jo boundary line hoti hai that is the representative
of the system).

String Conversion System.

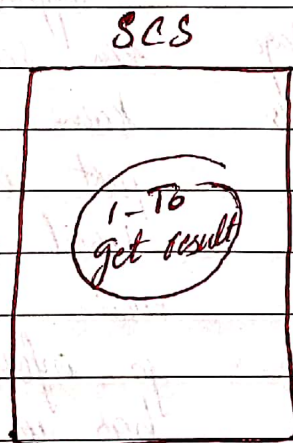
Applying UML

UML Pattern

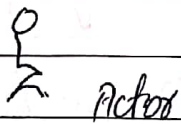
By Craig Larman

Ch 6, Ch 25

User



useCase → conceptual level py hota hai- & EBP → Physical (real time) level



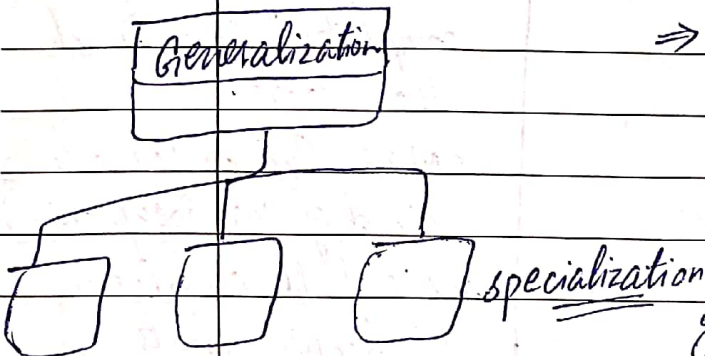
Relationship of actors if exists:-

Inheritance → Parent-child relationship

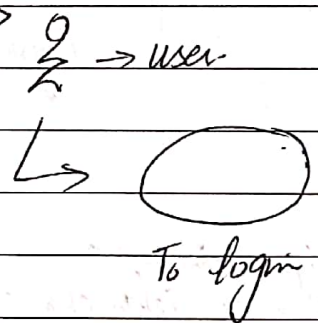
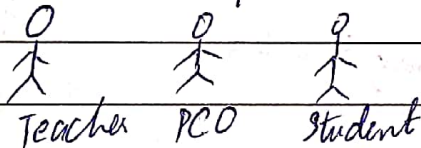
⇒ Properties or characteristics inherit hoty hai-

that relationship is used as generalization & specialization

generalization k baad specialization karta hai



In cms example aik actor hai user ab is se related sari properties use mai hogi- then specialization aye gi-



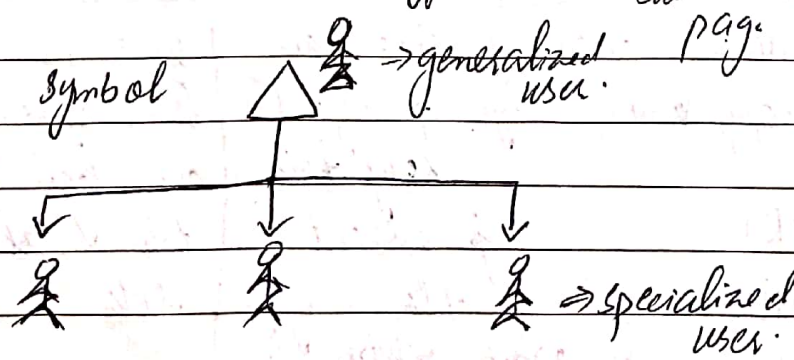
yahan py specialized properties a jayen gi-

specialized use cases honay sab k

To view student pag.

jahan zhusat pay gi wahan show karna hai.

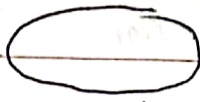
actors k relationships



These all are managed technically S → Specific
M → measurable

Date:/...../20.....

User



To login

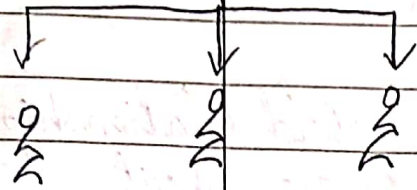
27 Nov 2023

exactly after 10 years.

Documentation

• Post → system completion hone baad ka state.

• Main Success Scenarios (kind of Mini Specifications) successful completion k scenario points likhna.



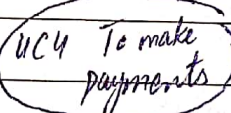
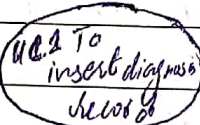
Student Teacher PCO



To view Student Page

⇒ Physician Billing System

Physician

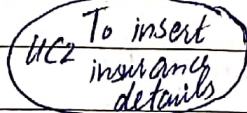
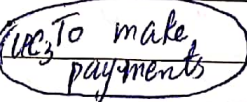


Insurance co

organizations ka vision jaldi change nahi hota but persons ka ↑

It is all about the future

Patient



ye teeno primary actors hai:

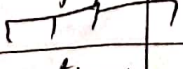
teeno main hai inka koi relationship

Vision → aik vision with time change ho skta hai.

Mission → ~~present~~ present → apni vision k bye ap kya kar rahi hai.

Goal

Objective



actions

SMART

↓ RULE

goals k sath wo apka mission hai.

wese nai ban saka lekin management k acc login krana jaise toh relationship ban skta-

↳ complete sentence ki abbreviation

Date:/...../20.....

(All day I dream about sports) 50+ sports Products.

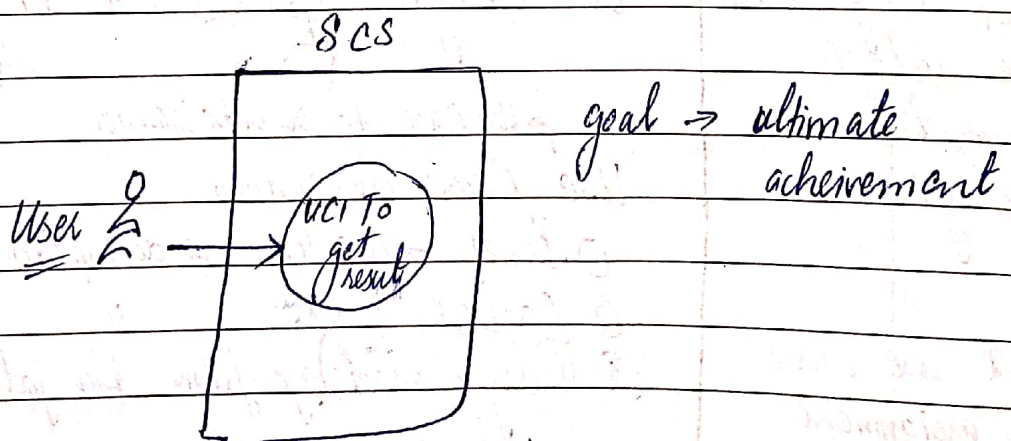
vision & mission is one sentence $\Rightarrow$ in se goal set hoty hai
has mission k apne apne goals hai
↳ multiple ho skty hai $\rightarrow$ has goal to achieve kame
k hye usk sub goals ya objective hongy.
jaise 8 semesters hmay objective hai-

goals ki sub division $\rightarrow$ objectives
or in objective k hye jo physical execution
ham kar rhy hai- to multiple objective
achieve hgy $\rightarrow$ then goal achieve hoga
↳ jaise goal hoga wese hi objective

tasks ko chod k baki sab kuch conceptual hai-
(actions)

us Use case diagram mai sary goals mil kar
system ka mission btayen gy- or vision bhi
(bottom to high OR high to bottom bhi
ho skti hai-

High level Use Case diagram $\rightarrow$ box use case hai-



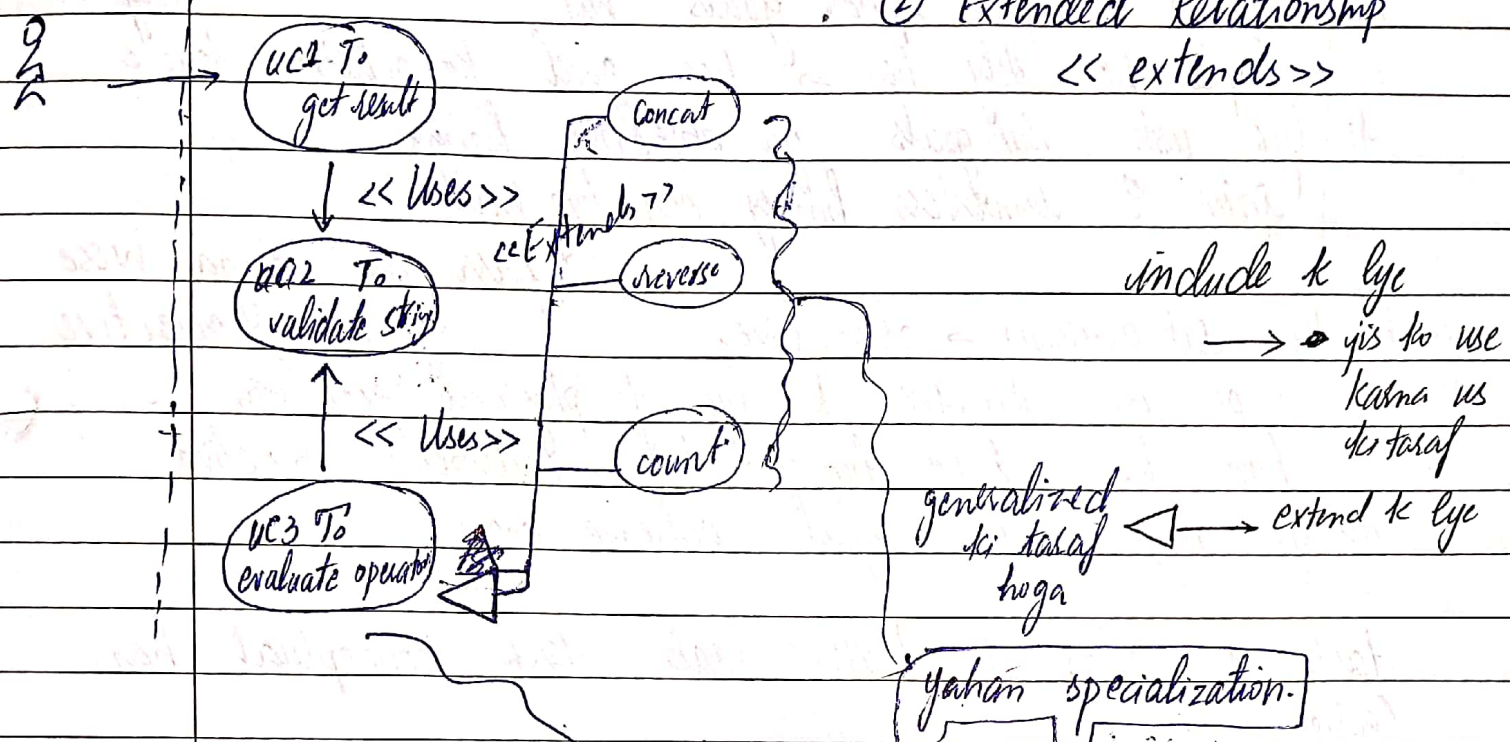
Relationship of UseCases → 2 types

- ① ~~include~~ relationship
 << Use >> → Stereotype
 ↓
 mandatory behaviour

Date:/...../20.....

① Include Relationship
 << Uses >>

② Extended Relationship
 << extends >>



<< Uses >> k andar behaviour mandatory hoga → k is k bagair kaam nahi ho sakti -

yahan generalization chal rahi hai

yahan py goals & subgoals hai inner details documentation mai aye gi -

jisko use karna hai arrow head us ki taraf. → include
 or triangle generalized ki taraf. ~~extension~~
 Use Case to describe karna.
 Use Case description:-

Six k use cases assignment banana quiz dena slides banana.

- ① Brief → Use case ki description only in one paragraph
 - ② Casual " in multiple paragraph.
 - ③ **Fully dressed** → ye follow kia jata bas ab
- From slides
 → environmental factors bhi a sakti hai is mai
 Pre conditions → koi specific kaam hone se pehly kya kya conditions true honi chahiye.

Software Engineering (SE)

Domain Model :-

→ the first in the object O Analysis where we take the concept of the classes

→ use case modeling to use karoni gy. classes ko identify karne k lie.

→ usko brane ka masqaad.

It is actually the way where we identify the classes

goal.

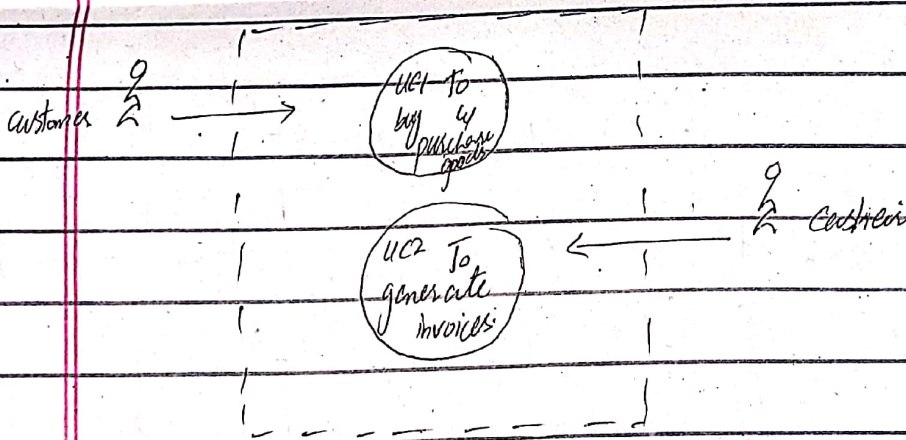
In ERD → Domain model (80-90%) same.

↓
entities to identify kia
than use

Our main focus is identification of classes & their relationships

kia tha
agy

Point of Sale System (High Level Use Case Diagram)



→ depends on situation

Two approaches to identify the classes.

→ Use Case Based Approach

class ke 3 parts

→ A customer jye honay chahiye
→ B items jye honay chahiye

Price of items is not the class. It is the attribute of the class.

Name
data member
functions or methods

nging

description
quantity price

ye sab attributes mai aye gy

C-
→ Invoice

→ D- Sale Person.

→ E Card

Use Cases to some rakh kr identify krna.

(2) Case Study Based Approach:-

Brainstorming

↳ Underline all the nouns (even they are duplicated)
↳ then make list of all the underlined words.

Focus
directed
thinking

Customer ✓ Checkout ✓ Basket ✓ Goods
Cashier ✓ Product ✓ Price ✓ Items
System ✓ Price ✓ Transaction ✓ Customer
Receipt ✓ Total ✓ customer ✓ Payment
credit Card ✓ Basket of Goods ✓ Card

inter related nouns ko alog kar dena hai-

Grouping
karni hai-

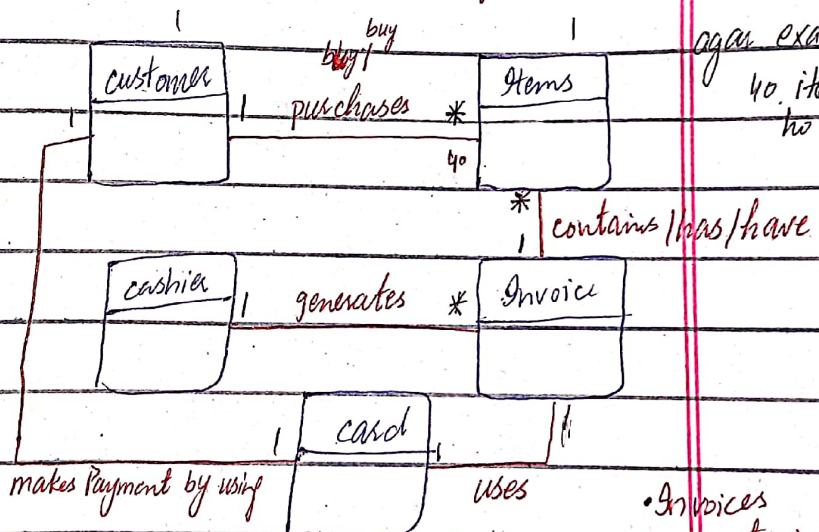
A- Customer

B- Cashier

C- Goods/Items

D- Card

E- Invoice



agar exact
40 items
ho

Relationship ko

association kaha jata

hai yahan.

↳ associate the

ERD mai relationship tha.

classes acc to . card uses

but yahan association

the requirements invoice to make payments.

present indefinite sentence

dena raha ho
simple ho-

1 → one
* → many

Identify

→ Association (ERD → Relationship)

→ Multiplicity (ERD → cardinality)

agar koi exact value ho to exact wahi likh

Items ki kam se kam 1 customer leni

Attribute → An attribute is defined by.

located)

records

is

res-

it

ai-

exact
no items
ho

we

is/has
have

ake

Domain model

mai functions or

methods ka hissa

available hota hai

lekin wo kahi hota hai

↓
or wo design
mai ya k

fill up hoga

• Nouns

• Remove duplications or same concept wali ko bhi
unique mai lana hai + irrelevant ko
bhi nikal dena hai.

eg system or basket ki class irrelevant

• conceptual class ko by k arrange

• then we declare relationships

domain model • We need to determine multiplicity of
classes

↳ all are conceptual classes

isi k parallel design class.

↳ jis ki waja se

Domain model is

also known as

conceptual design.

ye conceptual
nahi hogi.

ye software k

lehar se add ki jayen
gi.

isi ko braya.

gyc han

6-Dec-2023

SE

Wednesday

Lecture 10:-

Kya conceptual class ultimate solution hoti hai ya nahi?
⇒ conceptual classes is the integral part of the final solution

⇒ Object Oriented Design:- Interaction diagrams

Object design. { ① System Sequence Diagram → equivalent to Context level DFD
② Sequence Diagram
③ Collaboration Diagram -

Domain model mai methods nahi show karati wo is model mai hoti hai identify object design mai-

abstraction, refinement decomposition
ye sab yahan bhi hoga.

Use Case Model
↓
Domain model
↓
SSD

yahan bhi system black box hoga - or actors dikhaya gy but is mai aik sequence ana zoori hai-

Point of Sale System. Ica SSD (Domain model is the input for SSD)
↑
Use Case Model.

2 2

Sales Person Customer.

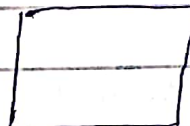
yahan pehly object ka name aye ga.

DFD mai sequence nahi hota.

SP1: Sales Person

C1 ~~SP1~~: Customer

has object of system
to neechy dotted lines

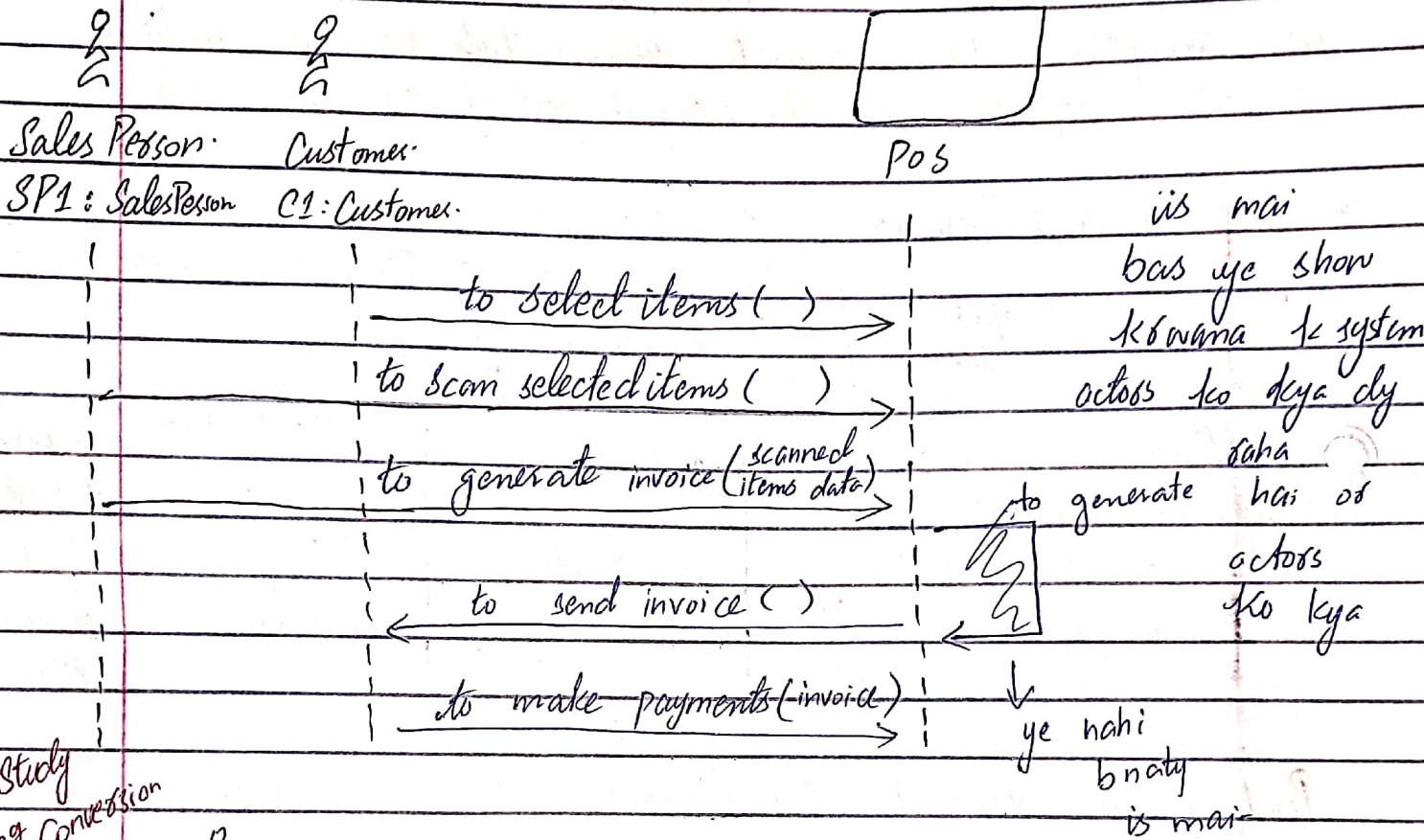


POS

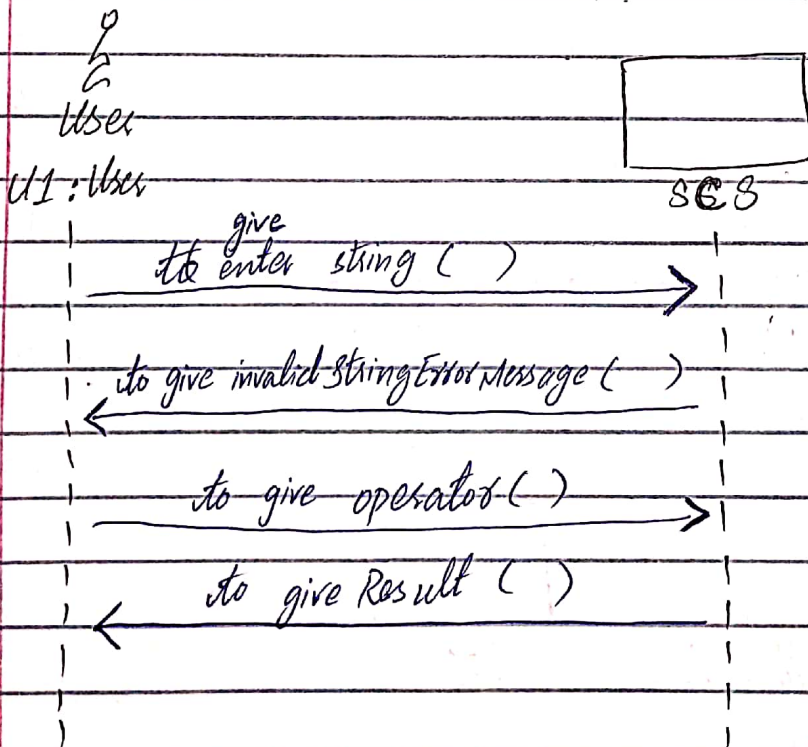
is called the
"life line of the object"

ye class purey system ko represent kar rahi hai-

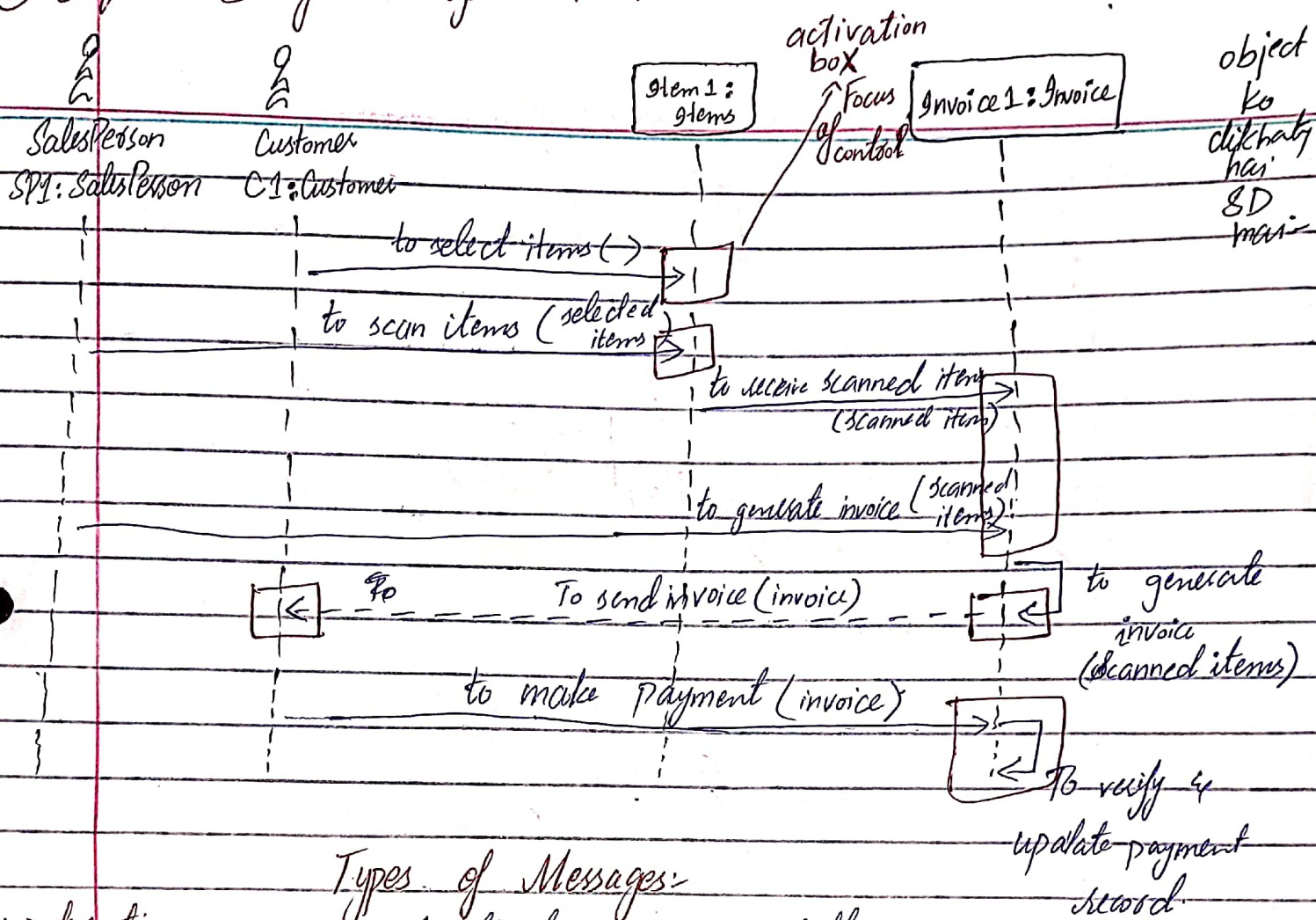
external entities jab message bhej sakti hai
 toh function call bhej sakti hai invoke kar
 sakti hai → function execution nahi hoti wo



Case Study
 String Conversion
 System



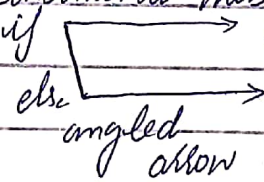
② Sequence Diagram:- system ko white box bana dena hai-



Types of Messages:-

inhi interaction
dikhani hai
respective object
ke sath.

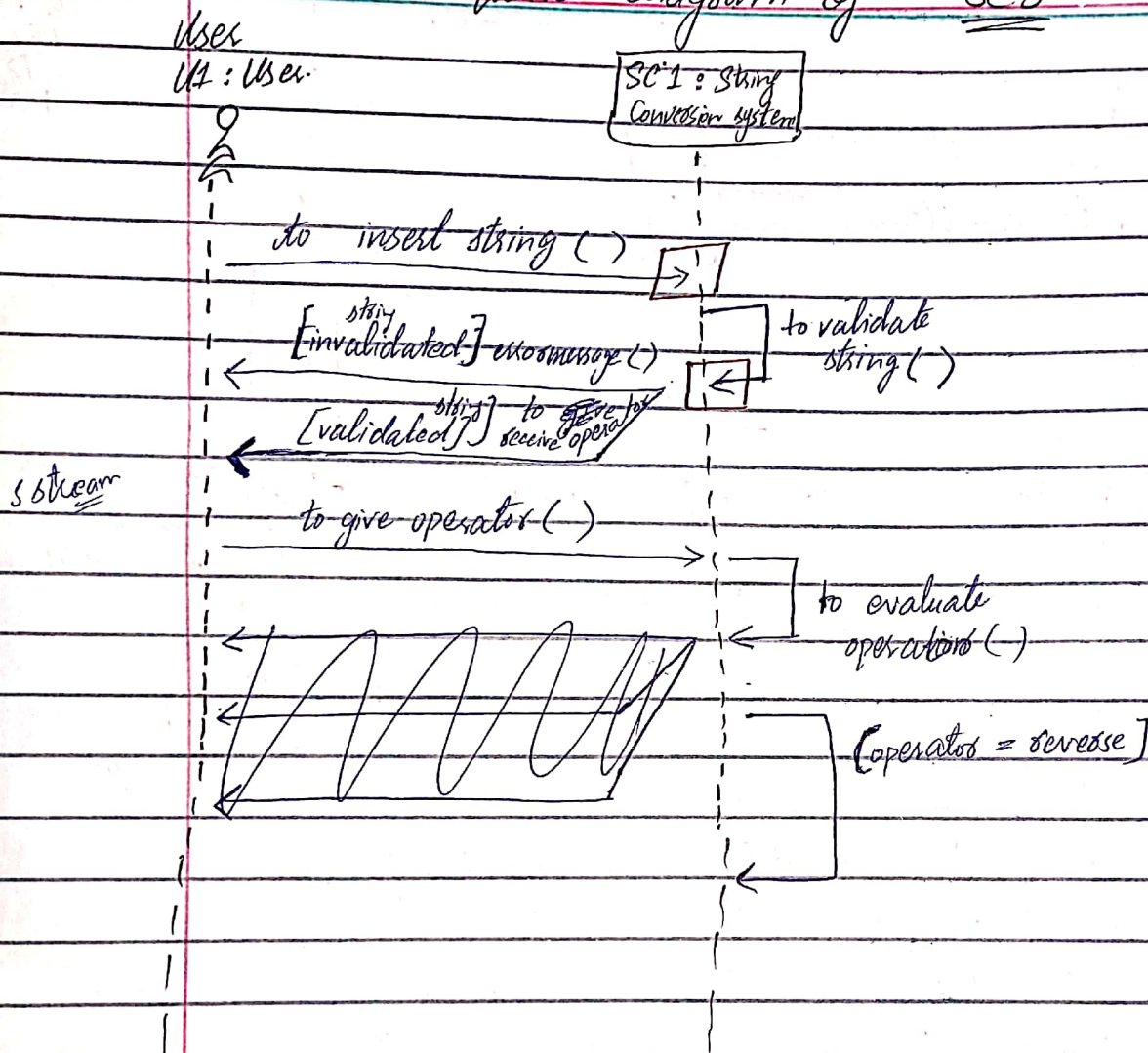
- 1 Simple message
- 2 Self message
- 3 Conditional message.
- 4 Mutually exclusive conditional message.
- 5 Object destruction message -
(by using <<destroy>> keyword).
- 6 Return message. (with dotted lines)



System Sequence mai actors or system k daungan thi interaction but

Sequence diagram of SCS

Sequence mai b/w objects / classes hongi

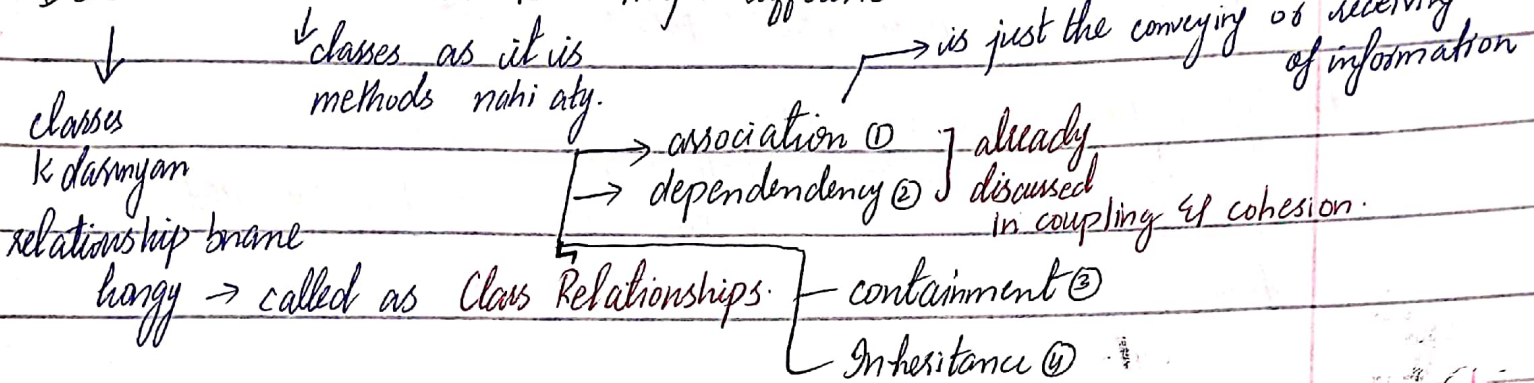


11 Dec 2020

SE

ODA k lehaaz se last diagram → Design Class Diagram.

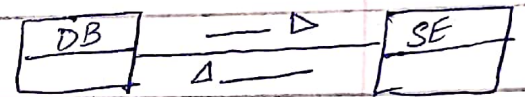
DCD or Domain model ka major difference:



⇒ dependency → is mai processing bhi hoti hai- association main sirf information ka liya (sirf) hota hai.

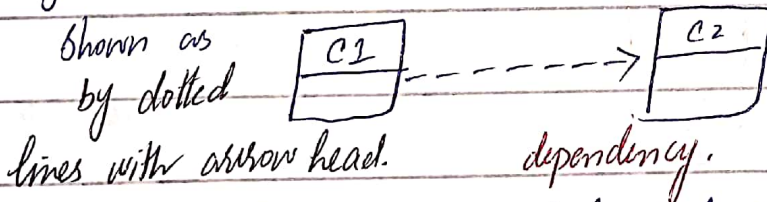
dependency → kisi function ko use karne ke liye kisi class ki zaroorat hai.

⇒ association is shown by straight line.



association ↓ Navigability arrow shows the direction of information.

in programming jab aik function k andar dusri function call ja rahi ho-



⇒ Containment → also known as (whole-part relationship)

in programming aggregation & composition.

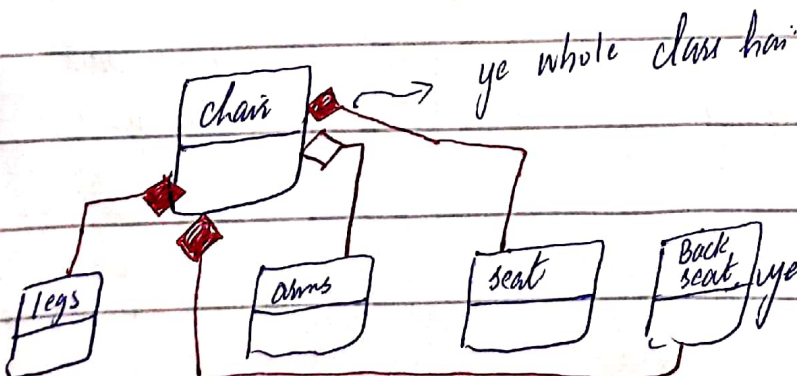
These two are types of containment

↳ whole class or part class k darmyan relationship.

this may be strong or weak relationship

arms k sath weak
back k sath strong

agar aise class k jany se dussri bhi khatam ho jaye.



new cheere kuch nahi

strong relationship → shown by diamond (filled) composition. | diamond ka face whole class ki taraf
 weak relationship → unfilled diamond → aggregation.

kisi dusri class ka object bana ke pass karo dena.

There are two ways to pass object

① by reference. → is se address pass hota hai → aggregation

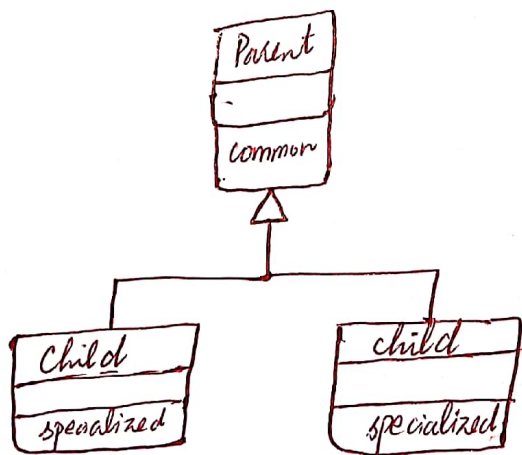
② by value. → This is composition.

↓
 is mai dono cheezin ja sahi hai-
 value bhi or reference bhi-
 strong relationship
 ↳ jo bhi cheez change hogi sari pay effect aye ga-

whole or parts ka damyan relationship.

Inheritance → specialized or generalized classes.
 ↓ child class ↓ parent class

Inheritance



There are 4 classes ka damyan proper relationships but data hiding, encapsulation or polymorphism. These are just concepts jo sab mai use ho sakti user management mai inheritance sab se zyada use hoti hai-

Polymorphism

↓
 ↳ some ~~same~~ diff name
 this is just a concept
 ↳ name diff behaviours
 diff ways of representing line
 ye kisi bhi cheez mai use ho skta hai-

DSA Quiz
 DB Quiz
 SE Assignment + Quiz
 Lab Quiz + Lab
 LA Quiz